

ΔCCORDION

Hack Days 2026: The AgentIQ Frontier

AUGUST 2026



The Business

Urban Media

A digital out-of-home advertising company operating ~11K digital screens across urban transit infrastructure in three major cities, which are deployed on buses, metro rail coaches, bus stops, metro platforms, and metro stations.

Each screen runs rotating digital ad slots that advertisers lease on a per-slot, per-day basis — from a single rotation to full-screen exclusivity. The company serves 500+ advertisers through a 50-person sales team, generating ~\$200M in annual revenue.

CURRENT CHALLENGES

Today, a sales rep gets an inquiry — an advertiser wants a 30-day campaign targeting commuters in the eastern metro corridor, budget ~\$50K. The rep opens a spreadsheet, eyeballs available screens, invents a price, and sends a proposal. No one knows if the price is right, if those screens fit the audience, or if three other advertisers are competing for the same inventory that week.



Inconsistent & Unguarded Pricing

Same screen quoted at different prices by different reps — no floor, no cap, no guardrails.



No Demand Optimization

Pricing ignores demand, occupancy, and seasonality — underselling high-value, oversitting low-demand.

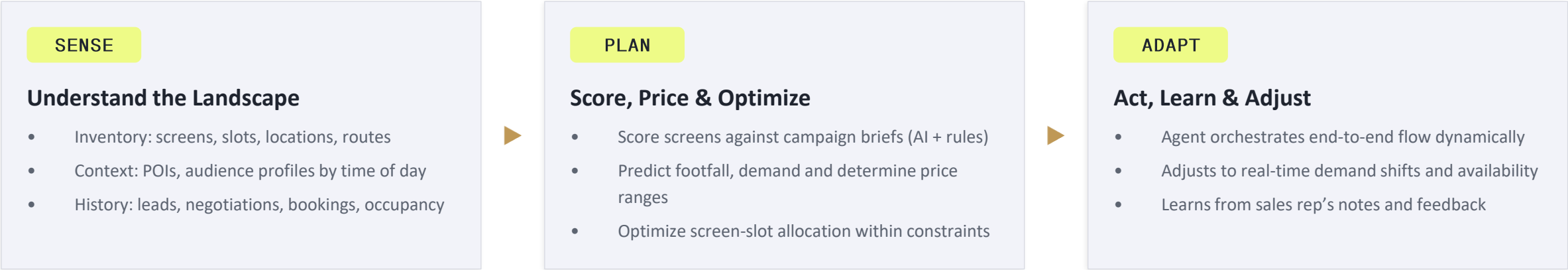


Blind Inventory Matching

No data-driven way to match advertisers to the right screens based on audience, budget, or goals.

Sense Plan Adapt

Build an AI-powered intelligence platform for the sales team that turns a messy campaign brief into the best-fit screens, right slots, optimal pricing, and a credible projected audience reach plus a clear reasoning for the choices



SOLUTION FLOW



Core Functionalities

Five functionalities. Each produces a concrete output that plugs into the solution flow.

Deliverable		Description	Techniques
D1	Audience Profile Engine	Takes screen locations, POI data, route/ridership data, and screen mount position to produce audience profiles for every screen — who is near this screen, when, and why. Leverage AI to infer profiles and apply proximity rules to filter noise.	AI Agent, Logical Rules
D2	Campaign–Screen Relevance Scorer	Takes a campaign brief and the audience profiles to produce a ranked list of screens with explainable relevance scores reflecting audience-campaign affinity — why this screen fits this campaign.	AI Agent, Logical Rules
D3	Demand Forecasting & Pricing Model	Uses historical bookings, lead pipeline signals, demand patterns, current availability, expected footfall human inputs to determine a price range (floor / target / cap) and a recommended optimal price per screen-slot in real-time.	AI Agent, Logical Rules, Forecasting
D4	Impressions Optimizer	Uses relevance scores, pricing, slot availability, and budget constraints to produce the optimal multi-location, multi-slot package that maximizes reach / impressions	Optimization Model
D5	Agentic Orchestration	Takes a raw campaign brief from the sales rep, orchestrates agents, and delivers a complete recommendation — screens, slots, pricing, projected impressions, and explanatory notes on why these choices were made.	Agent Orchestration
D6	Unified Platform	An interface for sales reps to interact with to upload relevant campaign documents, provide additional details, give feedback to agents and generate output	App development, UI/UX design

Dataset Overview: Five Layers

Fourteen tables, five layers — all the data Urban Media has to build audience profiles, relevance scores, and pricing is here.

	TABLES	TYPE OF INFO AVAILABLE
Geography	cities, zone_demographics, locations	City-level market tiers, zone demographics (income, age, density), and the physical catalog of bus stops and metro stations.
Network	route_stops, vehicles, route_schedules, ridership_actuals	Route topology and stop sequences, vehicle-to-route assignments, scheduled trips, and historical actual ridership.
Inventory	screens	The physical and vehicle-mounted screen catalog, including mount position and screen size.
Context	points_of_interest, events	Points of interest (type, footfall, proximity, side of road) and scheduled events (attendance, impact radius).
Commercial	client_facts, dim_slot, bookings, lost_leads	Advertiser profiles, time-block/slot definitions, historical booking transactions, and lost pipeline leads.

A few things worth thinking through

Not requirements — just the corners of the problem that tend to get overlooked as teams build.

Nuance	The subtlety
Scaling beyond these cities	Urban Media could expand to other locations and cities also in near future, with very different characteristics and demographics
Impressions are estimated	Ads with higher number of slots per minute have higher chances of being noticed than a quick 10s ad. Your recommendation should account for the non-linearity in impressions vs boards vs pricing
Demand is inferred	Build demand intensity from past leads, recent events in the areas, etc.
Explainability matters	A ranked list without reasoning isn't enough. Each recommendation should be able to say why — signals, POI fit, price rationale.
Some inventory has no history	Some screens might have sparse or zero historical data. Think about how your system behaves when signals are absent.
Boards on the same route share an audience	Two boards on the same route reach the same commuters. Treating them as independent reach is a modeling error worth handling.
A bundle isn't three independent decisions	A bus + metro + station package should be reasoned about as one deal — joint pricing and allocation, not three separate choices.
Not all demand signals age equally	Older unconverted leads are weaker signals than recent ones. The lead expiry date is in the data — consider how you weight it.
All other nuances you can think of.....	

What every team submits & how it'll be assessed

WHAT YOU'LL SUBMIT

1 Codebase & Environment

Full project source as a .zip, with a dependency manifest (venv / requirements.txt for Python, lockfile for Node.js). Includes a README with run instructions and a CLAUDE.md for AI coding context.

2 System Design — C4 Model

Context, Container, and Component diagrams delivered as a design doc, deck, html or chart (PDF or image export).

3 Demo Video

A 5–10 minute screen recording walking through the app's core features and capabilities in action.

HOW YOU'LL BE JUDGED

#	Criterion	What it measures
1	AI & Agentic Architecture	Soundness of how ML, OR, and agentic orchestration are combined into one coherent system.
2	Feature Depth & Purpose Fit	Breadth and depth of purpose driven features and capabilities to enhance the outcomes and experience
3	Recommendation Quality & Personalization	Degree to which outputs are tailored and audience-aware across campaign contexts.
4	Explainability & Trust	Clarity with which the system justifies its pricing and inventory recommendations.
5	Performance & Scalability	Ability to scale with inventory volume, balanced against latency vs. recommendation quality.
6	Code Quality & Engineering Rigor	Structure, readability, modularity, and reproducibility of the codebase.
7	Demo Impact & Communication	Clarity of the walkthrough in conveying the problem solved and value delivered.

Time Blocks & Rotation Slots

How one day of screen inventory is structured in dim_slot — and how a booking claims it

Each rotation slot isn't a chunk of the 4-hour block — all 6 play on a fast repeating loop (seconds each) for the entire block.

	time_block_id = 1 00:00-04:00 Night	time_block_id = 2 04:00-08:00 Morning	time_block_id = 3 08:00-12:00 Midday	time_block_id = 4 12:00-16:00 Afternoon	time_block_id = 5 16:00-20:00 Evening	time_block_id = 6 20:00-24:00 Night
Slot 1		Booking B			Booking A	
Slot 2					Booking A	
Slot 3					Booking A	
Slot 4						
Slot 5						
Slot 6						

Booking B (different client)
slots_booked_per_day = 1 ·
single_rotation

Booking A
slots_booked_per_day = 3
rotation_type = partial_rotation

Inside one time_block

1 > 2 > 3 > 4 > 5 > 6

↪ loops back to Slot 1

Slots cycle in seconds, not hours — the same 6 ads loop continuously for the full 4-hour block (hundreds of repeats/block).

RECURRENCE — the same slots recur every day across the booking's date range



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Unleash Your Agents!